

SOFIA AI ENGINEERING SPECIFICATION

Version 4.0

Phase 1 – AI System Specification

Part V – Professional Domain Framework

Status: APPROVED (Revision R1)

5.1 Purpose

Defines the Professional Domain Framework, describing how Sofia organizes, governs, and expands professional expertise across industries while maintaining one unified cognitive architecture.

5.2 Engineering Philosophy

Professional expertise is organized into domains rather than isolated assistants. All domains operate under one unified cognitive architecture.

5.3 Domain Architecture

Each domain includes: Domain Knowledge, Domain Workflows, Domain Documents, Domain Reasoning, and Domain Constraints.

5.4 Domain Lifecycle

Identify Domain → Define Knowledge → Map Professional Workflows → Catalog Documents → Configure Reasoning Rules → Apply Validation Standards → Integrate into Sofia Platform → Continuous Improvement.

5.5 Current Professional Domains

Executive Administration; Business & Strategy; Legal Support; Accounting & Financial Operations; Human Resources; Education; Healthcare Administration; Research & Analysis; AI Engineering; Content Development.

5.6 Multi-Domain Collaboration

Sofia dynamically combines expertise from multiple professional domains to address complex, cross-functional work.

5.7 Domain Independence

The platform is profession-independent. New domains may be added without redesigning the core architecture.

5.8 Domain Governance

Every domain follows Sofia's Cognitive Architecture, Professional Capability Framework, Human-Centered AI philosophy, and applicable legal, ethical, and regulatory considerations.

5.9 Future Domain Evolution

Future enhancements include organization-specific knowledge bases, workflows, templates, multilingual support, regulatory modules, validation engines, and enterprise integrations.

5.10 Engineering Summary

The Professional Domain Framework enables Sofia to organize and expand professional expertise through a unified engineering architecture while preserving consistency, scalability, and human-centered decision support.